Grazioso Salvare Rescue Dashboard

CS 340 Project Two README

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Project Purpose

The Grazioso Salvare Rescue Dashboard is a client-facing web application that connects the Austin Animal Center Outcomes data in MongoDB to an interactive Python dashboard. Its purpose is to help Grazioso Salvare identify and explore animals that match different search-and-rescue training profiles without requiring users to work directly with MongoDB commands. The dashboard provides interactive filtering controls, a responsive data table, a breed-distribution chart, and a geolocation map. A Grazioso Salvare logo and the unique identifier NSNYDER are displayed on the interface.

Required Functionality

The completed dashboard follows a model-view-controller approach. MongoDB serves as the model and stores the AAC shelter data. The reusable AnimalShelter CRUD Python module provides the database-access layer. Dash callbacks provide the controller logic that reacts to user selections, and the browser renders the interactive controls, data table, chart, and map as the view.

- Interactive rescue-type options allow the user to choose Water Rescue, Mountain or Wilderness Rescue, Disaster or Individual Tracking, or Reset.
- Each rescue-type selection triggers a MongoDB query through the existing CRUD Python module and refreshes the dashboard with matching records.
- The Dash DataTable supports pagination, sorting, filtering, and single-row selection.
- A Plotly pie chart summarizes the most common breeds in the current filtered selection.
- A Dash Leaflet map uses the selected table row to show the corresponding animal location and updates as the row selection changes.
- Reset restores the original unfiltered view of the Austin Animal Center Outcomes data.

How the Application Is Used

Run ProjectTwoDashboard.ipynb in Jupyter Lab. After the Dash application starts, open the generated Codio proxy link. The dashboard initially displays the full data set. Select one of the rescue-type radio buttons to filter the MongoDB data. The table and breed chart update automatically. Select a row in the table to move the marker on the geolocation map to that animal. Use Reset to return every widget to the unfiltered state.

Tools Used and Rationale

- **MongoDB:** Stores the Austin Animal Center Outcomes documents. Its document model works naturally with the shelter data and supports the query patterns needed by the rescue filters.

- **PyMongo:** Provides the Python driver used by the AnimalShelter CRUD class to connect to MongoDB and return query results to the dashboard.
- **Python:** Implements the reusable database module, data transformation, callbacks, and application logic.
- **Dash / JupyterDash:** Provides the web-application framework, interactive components, callback mechanism, and the connection between user controls and dashboard output.
- **Pandas:** Converts MongoDB query results into DataFrames and dictionaries that can be consumed by the Dash table and charts.
- **Plotly:** Creates the dynamic breed-distribution pie chart from the current filtered selection.
- **Dash Leaflet:** Creates the interactive geolocation map and marker based on the selected animal record.
- **Codio / Jupyter Lab:** Provides the hosted development environment used to run the notebook and deploy the Dash application through a proxy URL.

Useful resources: MongoDB Documentation | PyMongo Documentation | Dash Documentation | Plotly Documentation

Reproducing the Project

1. Start MongoDB and import aac_shelter_outcomes.csv into database aac and collection animals using mongoimport with CSV and header-line options.

2. Enable MongoDB authentication and create or verify the aacuser account with readWrite access to the aac database.

3. Place CRUD_Python_Module.py and ProjectTwoDashboard.ipynb in the same code_files directory. Keep the Grazioso Salvare logo image in that directory as well.

4. Install or verify the required Python packages: pymongo, pandas, dash, jupyter-dash, plotly, and dash-leaflet.

5. Enter the valid aacuser credentials in the dashboard code, then run ProjectTwoDashboard.ipynb from Jupyter Lab.

6. Open the generated Dash proxy link and verify the Reset, Water Rescue, Mountain or Wilderness Rescue, and Disaster or Individual Tracking states.

7. Confirm that the data table and breed chart change when a rescue filter is selected and that the map changes when a different table row is selected.

Challenges and Resolutions

Several implementation issues were resolved during development. MongoDB authentication had to be configured and verified so that the dashboard could retrieve data with the aacuser account. The CRUD Python module also required correction of Python indentation so that all class methods could be imported successfully. During dashboard development, an existing JupyterDash server thread caused a port-related error; restarting the kernel and using a fresh port resolved that issue. Finally, the browser view was zoomed out when documenting the finished dashboard so that the logo, unique identifier, filters, table, chart, and map could be captured together in each required screenshot.

Functional Test 1: Water Rescue

Grazioso Salvare Rescue Dashboard - NSNYDER

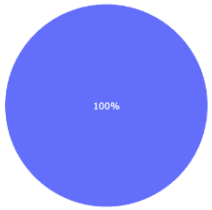
Rescue Type

○ Reset ◉ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster or Individual Tracking

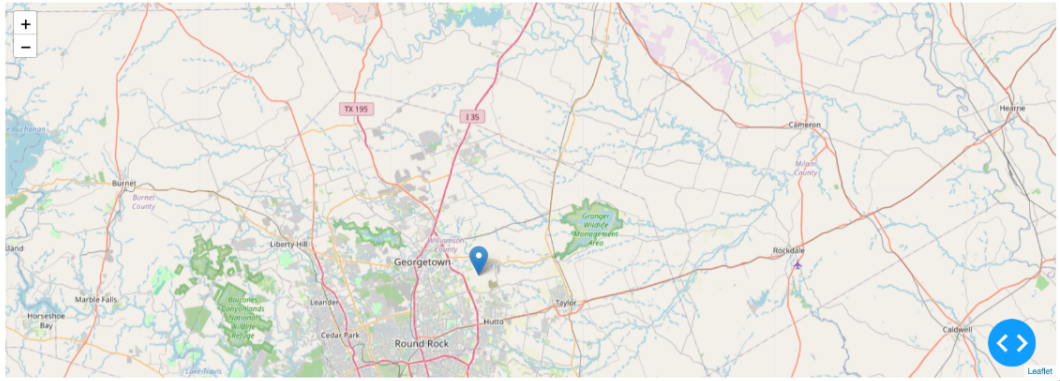
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	filter data...															
○	36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Euthanasia	Intact Female	30.5480802368633	-97.2969969058957	30.3544642857143
◉	732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14:59:00	2017-07-25T14:59:00	*Catalina		Return to Owner	Intact Female	30.6138310636757	-97.5752164857665	114.089186507937
○	1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14:12:00	2017-08-31T14:12:00	Pirata		Return to Owner	Intact Female	30.5572161697962	-97.5363224263878	52.3702380952381
○	1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Mika		Adoption	Intact Female	30.7569243032341	-97.7392549176654	40.2453373015873
○	1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15:20:00	2017-02-14T15:20:00	Marley		Return to Owner	Intact Female	30.4869754937324	-97.4280017197358	33.234126984127
○	1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:09:00	2017-12-03T13:09:00		Partner	Transfer	Intact Female	30.2840111162863	-97.4600542219677	53.0782738095238
○	2041	2 years	A702745	Dog	Labrador Retriever Mix	Black	2013-05-22	2015-05-22 11:45:00	2015-05-22T11:45:00	Abigail		Return to Owner	Intact Female	30.7157941956301	-97.4523664870572	104.355654761905
○	2225	2 years	A757341	Dog	Labrador Retriever Mix	Black/White	2015-09-01	2017-10-03 12:27:00	2017-10-03T12:27:00	19	Partner	Transfer	Intact Female	30.3814182796497	-97.7373217391863	109.074107142857
○	3319	9 months	A687748	Dog	Labrador Retriever Mix	Yellow	2013-12-09	2014-09-09 17:01:00	2014-09-09T17:01:00		Suffering	Euthanasia	Intact Female	30.729653440489	-97.3912941009824	39.2441468253968
○	4222	1 year	A735551	Dog	Labrador Retriever Mix	Black	2015-09-25	2016-09-27 14:10:00	2016-09-27T14:10:00	Daisy		Return to Owner	Intact Female	30.5150703656925	-97.6092589677489	52.655753968254

« ‹ 1 / 2 › »

Top Breeds in Current Selection



Labrador Retriever Mix



Water Rescue filter selected. The interactive table is narrowed to matching rescue candidates, and both the breed chart and geolocation map update from the filtered data.

Functional Test 2: Mountain or Wilderness Rescue

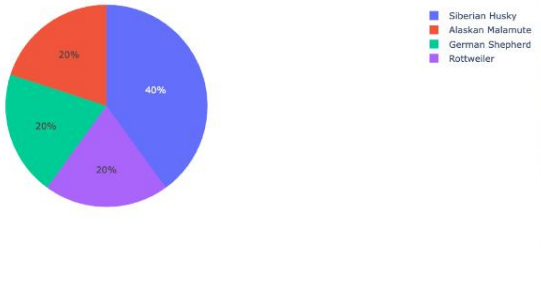
Grazioso Salvare Rescue Dashboard - NSNYDER

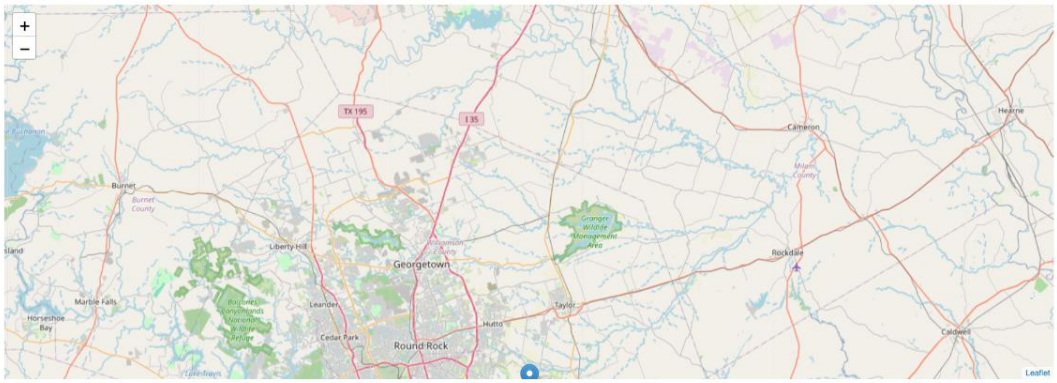
Rescue Type

○ Reset ○ Water Rescue ◉ Mountain or Wilderness Rescue ○ Disaster or Individual Tracking

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	filter data...															
◉	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24:00	2015-08-02T17:24:00	Papa		Return to Owner	Intact Male	30.4309339291938	-97.480825836737	104.817857142857
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male	30.40668985085	-97.485680334264	26.6422619047619
○	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00	2017-09-23T11:23:00	Zeke		Return to Owner	Intact Male	30.466577208743	-97.5573520930426	120.924900793651
○	3130	2 years	A721834	Dog	Siberian Husky	Brown/White	2014-03-05	2016-03-23 16:23:00	2016-03-23T16:23:00		Suffering	Euthanasia	Intact Male	30.5680998448899	-97.320550480325	107.09751984127
○	6191	2 years	A704101	Dog	Siberian Husky	Black/White	2013-06-01	2015-06-02 16:41:00	2015-06-02T16:41:00	Lobo		Return to Owner	Intact Male	30.4263764229275	-97.4309581796886	104.527876984127

Top Breeds in Current Selection





Mountain or Wilderness Rescue filter selected. The table, breed distribution chart, and map update together to reflect the selected rescue profile.

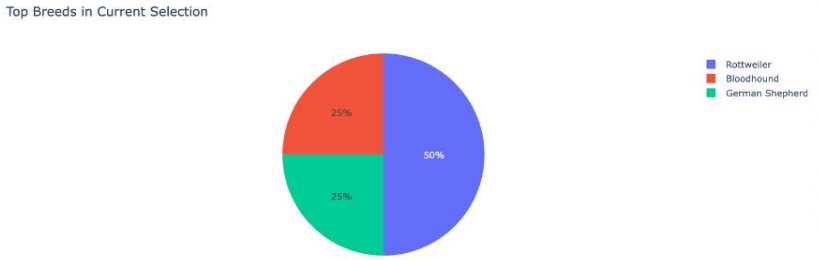
Functional Test 3: Disaster or Individual Tracking

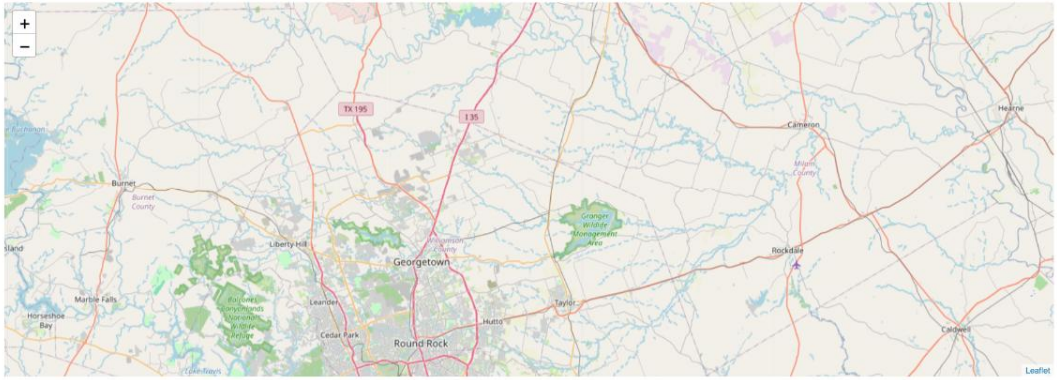
Grazioso Salvare Rescue Dashboard - NSNYDER

Rescue Type

○ Reset ○ Water Rescue ○ Mountain or Wilderness Rescue ◉ Disaster or Individual Tracking

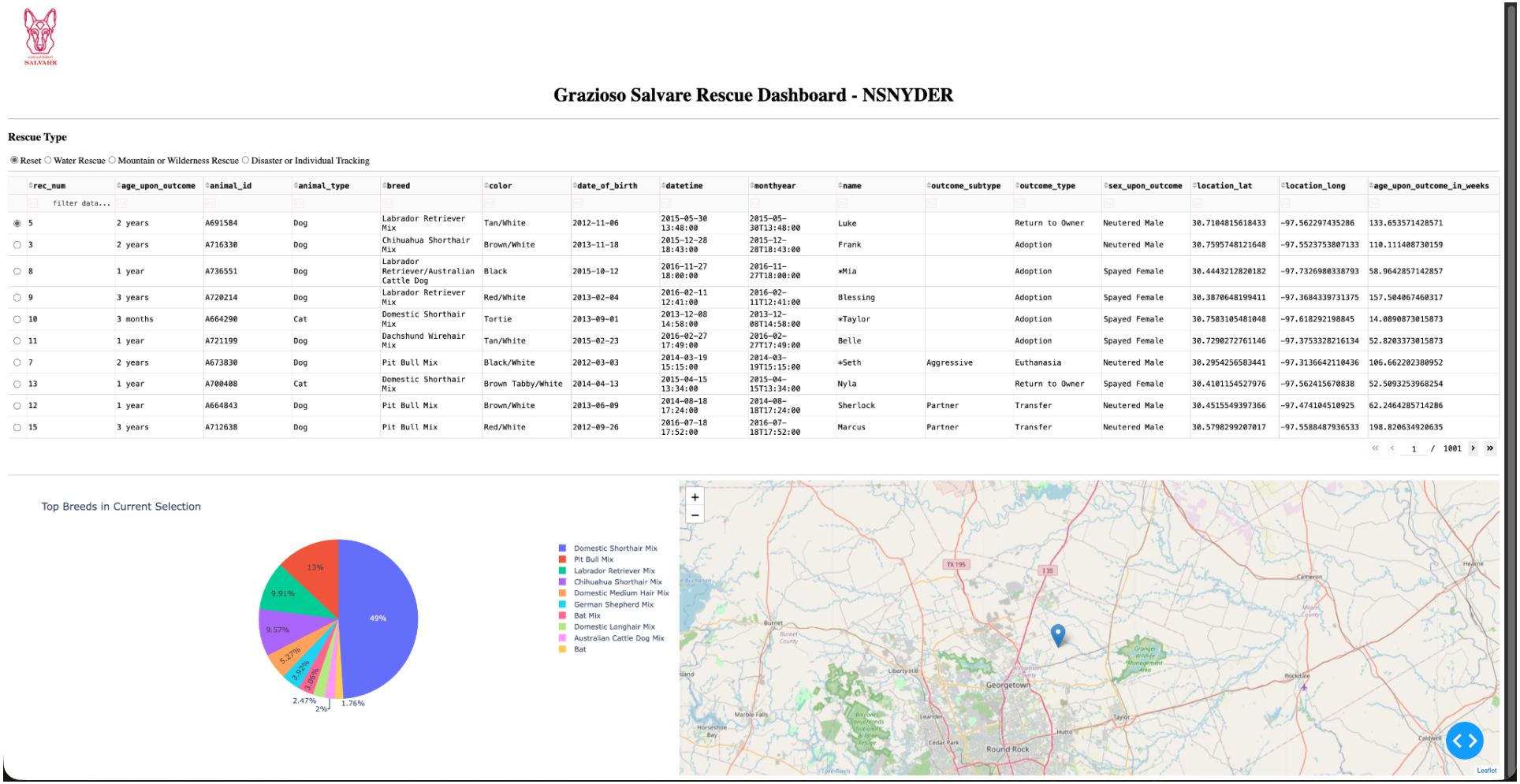
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	filter data...															
◉	3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43:00	2015-09-22T15:43:00	Boomer		Return to Owner	Intact Male	30.2709983761287	-97.5923916912722	209.093551587302
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male	30.40668985085	-97.485680334264	26.6422619047619
○	2987	4 years	A694614	Dog	Rottweiler	Black/Brown	2011-01-01	2015-01-01 14:25:00	2015-01-01T14:25:00	Striker		Return to Owner	Intact Male	30.329873203611	-97.5492968638502	208.800099206349
○	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00	2017-09-23T11:23:00	Zeke		Return to Owner	Intact Male	30.466577208743	-97.5573520930426	120.924900793651





Disaster or Individual Tracking filter selected. The dashboard dynamically refreshes the filtered table, chart, and map for the selected rescue profile.

Functional Test 4: Reset / Unfiltered State



	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	filter data...															
◉	5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke		Return to Owner	Neutered Male	30.7104815618433	-97.562297435286	133.653571428571
○	3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption	Neutered Male	30.7595748121648	-97.5523753807133	110.111408730159
○	8	1 year	A736551	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 18:00:00	2016-11-27T18:00:00	*Mia		Adoption	Spayed Female	30.4443212820182	-97.7326980338793	58.9642857142857
○	9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing		Adoption	Spayed Female	30.3870648199411	-97.3684339731375	157.504067460317
○	10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adoption	Spayed Female	30.7583105481048	-97.618292198845	14.0890873015873
○	11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spayed Female	30.7290272761146	-97.3753328216134	52.8203373015873
○	7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19T15:15:00	*Seth	Aggressive	Euthanasia	Neutered Male	30.2954256583441	-97.3136642110436	106.662202380952
○	13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown Tabby/White	2014-04-13	2015-04-15 13:34:00	2015-04-15T13:34:00	Nyla		Return to Owner	Spayed Female	30.4101154527976	-97.562415670838	52.5093253968254
○	12	1 year	A664843	Dog	Pit Bull Mix	Brown/White	2013-06-09	2014-08-18 17:24:00	2014-08-18T17:24:00	Sherlock	Partner	Transfer	Neutered Male	30.4515549397366	-97.474104510925	62.2464285714286
○	15	3 years	A712638	Dog	Pit Bull Mix	Red/White	2012-09-26	2016-07-18 17:52:00	2016-07-18T17:52:00	Marcus	Partner	Transfer	Neutered Male	30.5798299207017	-97.5588487936533	198.820634920635

« ‹ 1 / 1001 › »

Reset selected. The dashboard returns to the original unfiltered Austin Animal Center data while retaining the interactive table, breed chart, and geolocation map.